

PIONEER JUNIOR COLLEGE
JC2 PRELIMINARY EXAMINATIONS 2011

**Candidate
Name**

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**Civics
Group**

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**INDEX
NUMBER**

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H2 BIOLOGY

9648 / 02

PAPER 2 CORE PAPER

13 Sept 2011

2 hours

READ THESE INSTRUCTIONS FIRST

Write your name, class and index number on all the work you hand in.
Write in dark blue or black pen.

You may use a soft pencil for any diagrams, graphs or rough working.
Do not use staples, paper clips, highlighters, glue or correction fluid.

Read the instructions on the answer sheet very carefully.

This document consists of 2 sections:

Section A

Answer **ALL** questions.

Answers are to be written in the spaces provided.

Section B

Answer only **ONE** question.

Write your answer on the foolscap paper provided.

Do not open this booklet until you are told to do so.

For Examiner's Use	
Section A	
1	/ 10
2	/ 12
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Section B	
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TOTAL	/ 100

This document consists of **23** printed pages (inclusive of this page)

[Turn Over

- 1 In an experiment to study the effect of heat treatment on the digestibility of protein substrate and the effect of raw bean extract on protease activity, various reaction mixtures were prepared and were incubated for 30 minutes.

The protein concentration of each reaction mixture at the beginning and at the end of the incubation period was determined by the colorimetric method which measures colour intensity of these reaction mixtures. The results were shown in table 2.1.

Table 2.1

Incubation period /min	Colour intensity of the reaction mixture / arbitrary unit		
	Tube A	Tube B	Tube C
	Protease + heated protein substrate	Protease + unheated protein substrate	Protease + unheated protein substrate + raw bean extract
0	10	10	10
30	4	7	9

Fig. 2.1 shows a standard graph obtained by using colorimetric method for determining concentration of protein solutions.

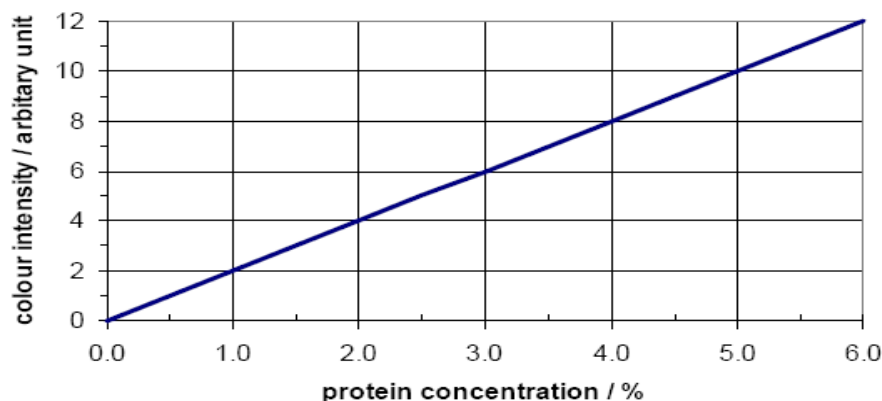


Fig 1 Fig. 2.1

(a) With reference to Table 2.1 and Fig 2.1,

- (i) Determine the protein concentration of the three tubes at time 0 min.

5% protein concentration; @1 mk [1]

- (ii) Explain which protein substrate is more digestible by protease.

heated protein more digestible by protease;

less protein remained after incubation with protease / lowest colour intensity;

unheated = 3.5% protein concentration / decreases by 1.5%;
vs heated protein = 2% / decreases by 3% ; @½ mk [2]

(b) Based on your knowledge of protein structure, explain the effect of heat treatment in the digestibility of this protein substrate.

- **heating increases the digestibility of this protein substrate / digest more quickly / easier to digest;**

-
- **heating causes protein substrate denaturation**

-
- **Ref. to easier access of enzyme(s' active site) to substrate/peptide bonds;**

[3]

(c) Suggest the effect of raw bean extract on protease activity.

inhibits protease activity/competitive/non-competitive inhibitor;

R: inhibitor to substrate

@ 1 mk

[1]

(d) Fig 2.2 shows the structure of a biochemical molecule, **W**.

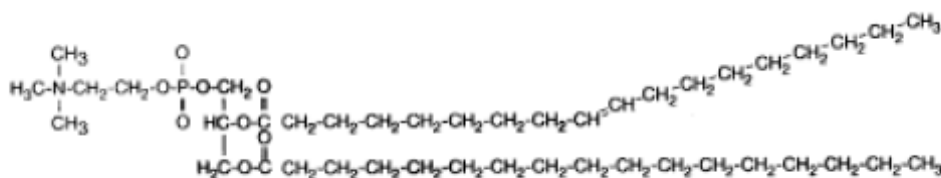


Fig ' Fig 2.2

(i) Identify the molecule **W**.

phospholipid;

@1mk

[1]

(ii) Membrane lipids in reindeer hooves need to remain fluid at low temperatures. Based on the structure of molecule **W**, predict the composition of such membrane lipids.

high proportion of phospholipids with unsaturated fatty acids tails/

high proportion of cholesterol;

@ 1mk

[1]

(iv) The feather of certain aquatic birds such as ducks are covered with a protective layer of wax. Based on this information, predict why oil spills are so harmful to these birds.

oil dissolves protective wax of feathers;

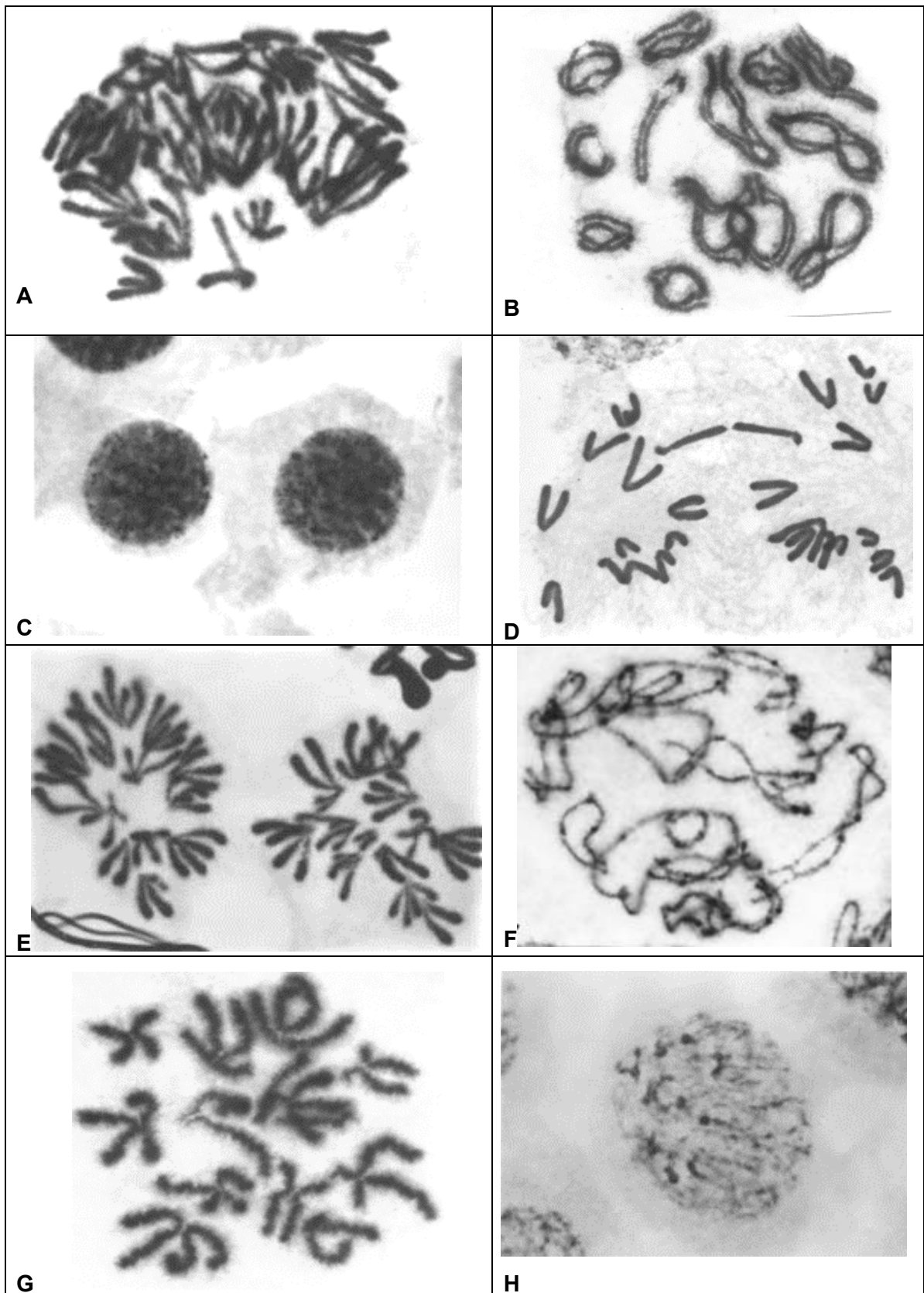
buoyancy lost/poor insulation in cold/ waterproofing lost/bird cannot keep warm;

R: oil causing feathers to stick without ref. to wax

[Total: 12]

Question 2 NJC 2008

The micrographs below show nuclei of cells at various stages during nuclear division in a flowering plant.



(a) Micrographs **C** and **H** show the 5th and 6th stages in the sequence. In the boxes below, arrange the other letters shown on the micrographs to indicate the correct

chronological sequence for this type of nuclear division. [2]

F	B	A	E	C	H	G	D
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(b) What is the diploid number of this plant? With reference to at least 1 stage from the micrographs, explain how you arrive at your answer. . [max 2]

Diploid number of this plant = 26
Any combination of 2 micrographs/ stages B, D, E or G, with explanations. Stage B (Prophase 1); 13 pairs of homologous chromosomes/bivalents; already paired up → total of 26 chromosomes;
Stage G (metaphase II); Aerial view of 13 chromosomes - haploid no.; (each with 2 chromatids) lined up along equator ;
Stage D (anaphase II); 13 separated chromatids (single strands); at each pole ;
Stage E (anaphase I) ; 13 separated chromosomes at each pole; each with a centromere;

(iv) Discuss the significance of stage B [2]
At B, prophase I

- pairing up of homologous chromosomes via synapsis which later allows for the alignment of chromosomes at metaphase 1.
- This will allow separation of homologous chromosomes and halving of chromosome number so that during fertilisation of gametes (no doubling/restoration of diploid number.)
- Non-sister chromatids of homologous chromosomes twist around each other making contact at several points along their length- chiasmata.
- At each chiasma, exchange of chromosomal material between non-sis chromatids. → crossing over and exchange of alleles → Leads to genetic variations among the 4 haploid cells.

(d) Figure 2.3 shows changes in DNA content at different stages of the animal life cycle.

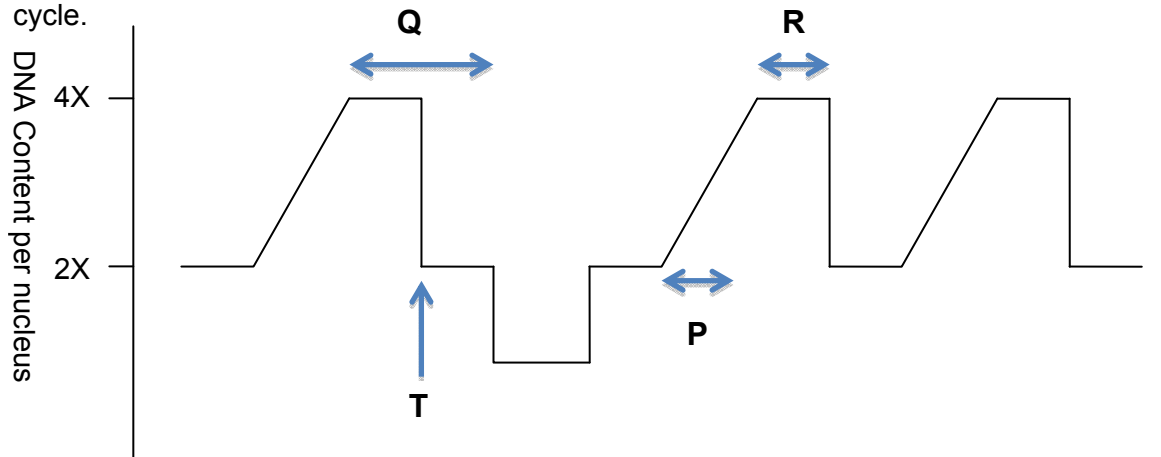


Figure 2.3

With reference to Figure 2.3,

(i) identify and describe Stage P. [2]

- **DNA replication during Interphase**
- **Doubling of DNA content from 2X to 4X**
- **2 parental DNA strands act as templates for synthesis of complementary daughter strands**
- **DNA polymerase; new strand formed in 5' to 3' direction; proofreading**

(ii) Identify the type of division and justify your answer. [2]

R

- **Mitosis**
- **Started off with 2x DNA content before P where DNA replication occurs during S phase. After R takes place, amt of DNA in each daughter nucleus is the same as the parent before P process;**

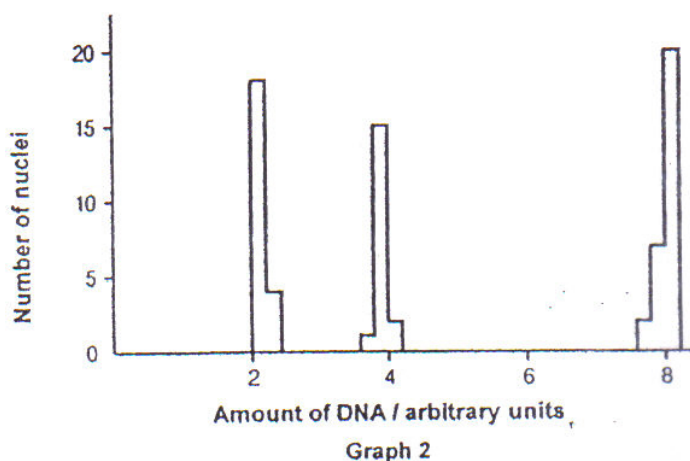
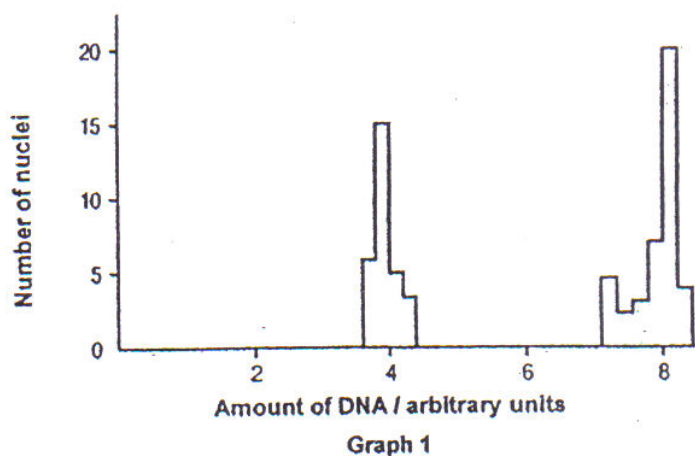
Q

- **Meiosis**
Either:
- **Started off with 2X amt of DNA by the end of Q, the amount of DNA in each daughter nucleus is halved → X amt**
Or:
- **There are two nuclear divisions seen: once at T and the second when the DNA was halved.**

(iii) Identify stage T. [1]

Telophase I

The formation of sperm cells involves mitosis and meiosis. The graphs below show the amount of DNA in the nuclei of cells taken from different parts of a mammalian testis.



With reference to the graphs,

(e) (i) Explain the evidence which shows that Graph 1 represents cells undergoing mitosis. [1]

- Two peaks (4 and 8 DNA arbitrary units) representing DNA replication and cytokinesis after one nuclear division.
- Absence of a 3rd peak/ group of cells with ~2 DNA arbitrary units.

(ii) Explain how the events of meiosis result in three groups of nuclei in Graph 2. [2]

- Parent cells have ~3.5-4.2 (range) units of DNA in 1st group of cells, cells undergo DNA replication in interphase I
- DNA replication occurs during S phase of interphase → cells have twice the amount of DNA as compared to parent cells → 7.5 – 8.2 units of DNA
- Homologous chromosomes separate during anaphase I, thus cells would have undergone reduction division ie half the no. of chromosomes as compared to Diploid no. → cells have ~3.5-4.2 (range) units of DNA.
- after telophase II, sister chromatids separate during anaphase II, the haploid cells will have half the amount of DNA → 2 – 2.5 units of DNA

[Total: 12 marks]

3(a) Explain how the promoter of a eukaryotic gene influences transcription. [2]

- **It is a regulatory non-coding DNA sequence (found at 5' end of a gene)**
- **allows binding of general transcription factors to TATA box**
- **allows binding of RNA polymerase (II)**
- **to form transcription initiation complex**
- **allows for initiation of transcription → determines which gene gets expressed or not**

(b) Fig. 3.1 shows the binding of a general transcription factor to part of the promoter. This will help to recruit RNA polymerase to the promoter.

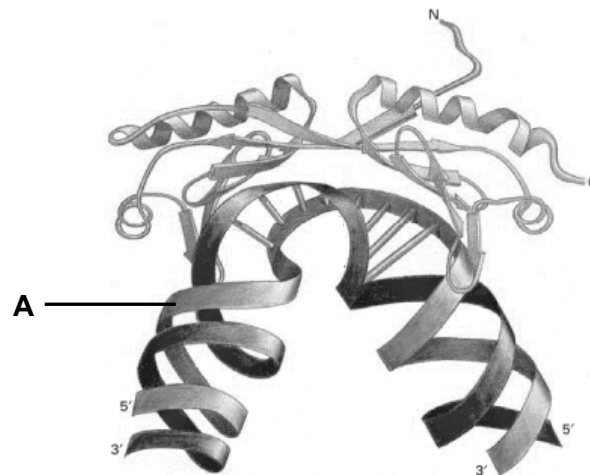


Fig. 3.1

(i) With reference to Fig. 3.1, give **one** reasons why the structure labelled **A** is part of the promoter and not the transcription factor. [1]

- **Promoter is made of DNA → double helix / stranded**
- **Has 5' and 3' ends which run antiparallel**

(ii) Suggest why a substitution mutation at the promoter may result in the silencing of the gene controlled by the promoter. [1]

- **Substitution mutation changes base sequence at one or a few bases so new sequence is no longer complementary to binding site of / not recognised by transcription factors / RNA polymerase**
- **Transcription factors cannot bind and cannot mediate binding of RNA polymerase to promoter → transcription cannot be initiated and is switched off.**

(c) Describe how distal control elements can help increase the rate of transcription of a gene. [2]

- **Activators bind to distal control elements / enhancer**
- **Recruits DNA bending protein/causes DNA to bend**
- **More transcription factors and RNA polymerase brought to promoter**
- **Form the transcription initiation complex**

- (d) Even though the control of gene expression occurs mostly at the level of transcription, control of gene expression also occurs at the level of post-translational modification

Name an example of post-translational modification of a **specific** protein [1]

- **eg. insulin; cleavage; / histone; acetylation/methylation**
- **protein kinase; phosphorylation**
- **ubiquitination (no specific eg for this); or any other valid eg**

Human DNA has been analyzed and details of certain genes are shown in the table below.

Gene	Gene size / kb*	mRNA size / kb	Number of introns
Insulin	1.7	0.4	2
Collagen	38.0	5.0	50
Albumin	25.0	2.1	14
Phenylalanine hydroxylase	90.0	2.4	12
Dystrophin	2000.0	17.0	50

*Kilobase pairs

(Source: William S Klug and Michael R Cummings, (2002), Concepts of Genetics, 7th Edition, Prentice Hall, page 314)

- (e) (i) Calculate the average size of the introns for the albumin gene. (Show your workings.) [1]

- **$25 - 2.1 = 22.9\text{kb}$;**
- **$22.9/14 = 1.6\text{kb}$;**

- (ii) Briefly discuss the relationship between gene size and the number of introns. [1]

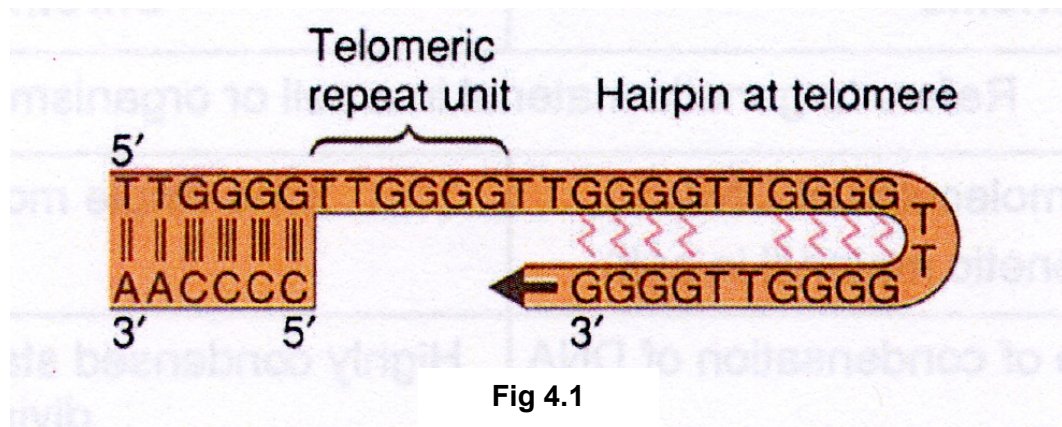
- **Generally, the bigger the gene size, the higher the number of introns;**
- **Ref. to insulin gene (1.7kb) with 2 introns and collagen gene (38kb) with 50 introns;**
- **The general trend has exceptions;**
- **Ref. to phenylalanine hydroxylase or dystrophin genes;**

- (iii) Determine the maximum number of amino acids that could be produced by translating the phenylalanine hydroxylase mRNA. (Show your workings.) [1]

- **$2400 - 3 = 2397$; (3 bases for the stop codon)**
- **$2397/3 = 799$ amino acids;**

[Total: 10 marks]

4. **Figure 4.1** shows a diagram of a telomere. Telomeres possess a unique single stranded G-rich 3' overhang that loops back to form a hairpin structure as shown.



- (a) Describe the role of telomeres [1]
- Repetitive sequence found at the end of a chromosome.
 - Plays a protective function because it delays degradation of genes near the end of the telomeres. OR
 - Prevent ends of chromosomes from fusing which makes a genome unstable; OR
 - Provide counting mechanism for cells to undergo apoptosis when the telomere becomes too short.
- (b) The telomerase enzyme is responsible for extending the telomere. It is essential in single celled organisms, while in humans and other vertebrates, it is active in reproductive cells.
- (i) Explain how the 3' overhang is formed at the end of a linear chromosome [1]
- Primer removed but cannot be replaced
 - Resulting in a 3' SS DNA
- (ii) Explain why regeneration of telomeres is not normally required in human body cells. [1]
- Most human cells required to divide a limited number of times
 - After which the cell undergoes apoptosis/programme cell death when the telomeres are too short.

(d) The figure 4.2 below shows the sequence of events in a Ras cell cycle-stimulating pathway.

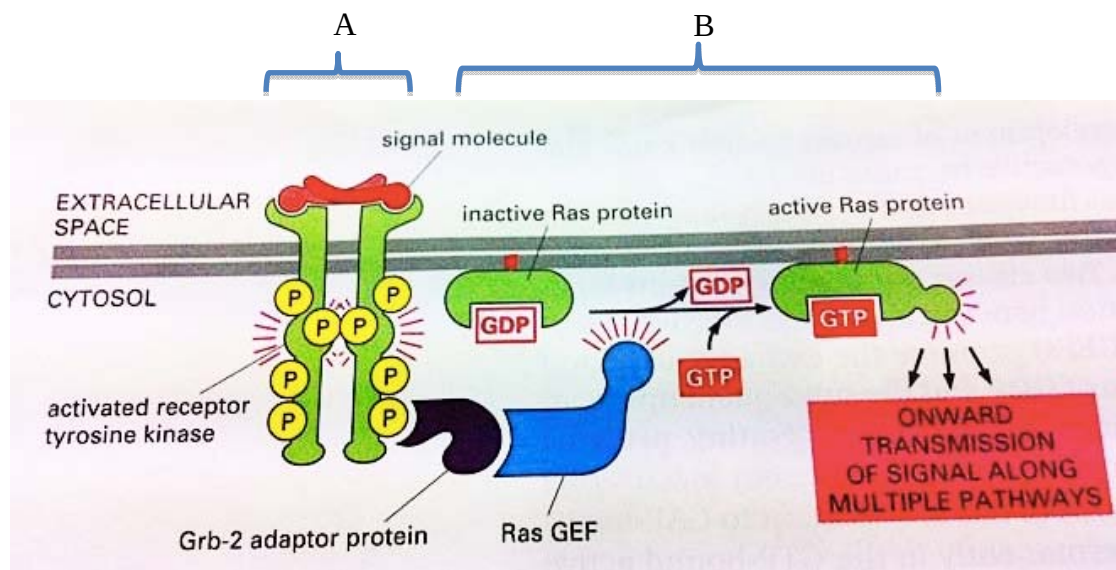


Fig. 4.2

(i) Briefly describe the stages A and B shown in Fig. 4.2. [3]

A

- **Dimerization of the 2 receptors;**
- **activation** of the receptor tyrosine kinase region leads to **phosphorylation** of tyrosine kinases of the other
- followed by subsequent phosphorylation of tyrosine amino acids on the tail of subunits;

B

- Grb-2 adaptor protein recognizes and **binds** to specific **phosphorylated tyrosine(s)**
- In turn binds Ras GEF which becomes **activated**
- which activates Ras protein by removing GDP and replacing it with GTP in order to activate Ras protein to bring about

(ii) Identify the signal molecule and explain why it cannot act directly on the DNA in the nucleus. [2]

- **Growth factor**
- **It is probably hydrophilic, possibly polar or charged**
- **Or due to its large size**

(iii) Explain how a small number of signal molecules can bring about a large cellular response. [2]

☺ = **Signal amplification**

☺ **At each catalytic step in the cascade, the number of activated products is much greater than in the preceding step.**

☺ **The amplification effect stems from the fact that these proteins persist in the *active form* long enough to process numerous molecules of substrate before they become inactive again.**

(iv) With reference to Fig. 4.2, suggest how mutation in the Ras gene may bring about cancer. [2]

- **Ras gene mutated**
- **→ hyperactive ras mutation**
- **Can no longer be turned off even when ligand/growth factor is no longer bound to the receptor tyrosine complexes**
- **Continuously stimulate cell division continuously and are no longer density and anchorage dependent → tumor**

[Total: 12 marks]

QUESTION 5

(a) Define what is meant by a pure-bred plant.[1]

A plant that is homozygous at all gene loci OR produces only offspring of the same variety when it is self-pollinated.

(b) Using suitable symbols, draw a genetic diagram to explain the results of the test cross. [5]

Let A represent the dominant allele for green leaves, a represent the recessive allele for mottled leaves.

B represent the dominant allele for smooth fruit, b represent the recessive allele for hairy fruit.

F1 test cross Green leaves, smooth fruit x Mottled leaves, hairy fruit

	$\frac{A\ b}{a\ B}$	$\frac{a\ b}{a\ b}$													
Gametes	$\frac{A\ b}{a\ B}$ $\frac{A\ B}{a\ b}$	$\frac{a\ b}{a\ b}$													
Random Fertilization	<table><tr><td colspan="4">Female Gametes</td></tr><tr><td>$\frac{A\ b}{a\ B}$</td><td>$\frac{a\ B}{a\ b}$</td><td>$\frac{A\ B}{a\ b}$</td><td>$\frac{a\ b}{a\ b}$</td></tr><tr><td>Male Gametes $\frac{a\ b}{a\ b}$</td><td>$\frac{A\ b}{a\ b}$</td><td>$\frac{a\ B}{a\ b}$</td><td>$\frac{A\ B}{a\ b}$</td><td>$\frac{a\ b}{a\ b}$</td></tr></table>		Female Gametes				$\frac{A\ b}{a\ B}$	$\frac{a\ B}{a\ b}$	$\frac{A\ B}{a\ b}$	$\frac{a\ b}{a\ b}$	Male Gametes $\frac{a\ b}{a\ b}$	$\frac{A\ b}{a\ b}$	$\frac{a\ B}{a\ b}$	$\frac{A\ B}{a\ b}$	$\frac{a\ b}{a\ b}$
Female Gametes															
$\frac{A\ b}{a\ B}$	$\frac{a\ B}{a\ b}$	$\frac{A\ B}{a\ b}$	$\frac{a\ b}{a\ b}$												
Male Gametes $\frac{a\ b}{a\ b}$	$\frac{A\ b}{a\ b}$	$\frac{a\ B}{a\ b}$	$\frac{A\ B}{a\ b}$	$\frac{a\ b}{a\ b}$											
Phenotypes	Green leaves, hairy fruit 46	Mottled leaves, smooth fruit 40	Green leaves, smooth fruit 8	Mottled leaves, hairy fruit 6											
	Parental		Recombinants												

(c) Explain the deviation of the observed test cross results from the expected dihybrid test cross ratio. [2]

- Expected dihybrid test cross ratio is 1:1:1:1 while observed test cross results do not show the ratio
- as parental phenotypes occur in higher frequency than recombinant frequencies which indicates that the 2 genes for leaf colour and fruit texture are linked.

(d)(i) This second scientist claimed that the observed test cross ratio of offspring can be simplified to 1:1. Perform the chi-squared (χ^2) test to verify his claim. [2]

$$\chi^2_{\text{calculated}} = [(52-48)^2 + (44-48)^2]/48 = 0.67$$

At $p = 0.05$, degree of freedom = 1, $\chi^2_{\text{critical}} (0.05, df=1) = 3.84$

Since $\chi^2_{\text{calculated}} < \chi^2_{\text{critical}}$, and probability of $\chi^2_{\text{calculated}}$ is $>$ than $P = 0.05$, there is no significant difference between observed test cross ratio and expected 1:1 ratio. Hence, the scientist's claim is valid

Question 6

Dopamine is a chemical naturally produced in the body. In the brain, dopamine functions as a neurotransmitter, activating dopamine receptors on the post-synaptic membrane. Dopamine concentration affects brain processes that control body movement, emotional response and the ability to experience pleasure and pain.

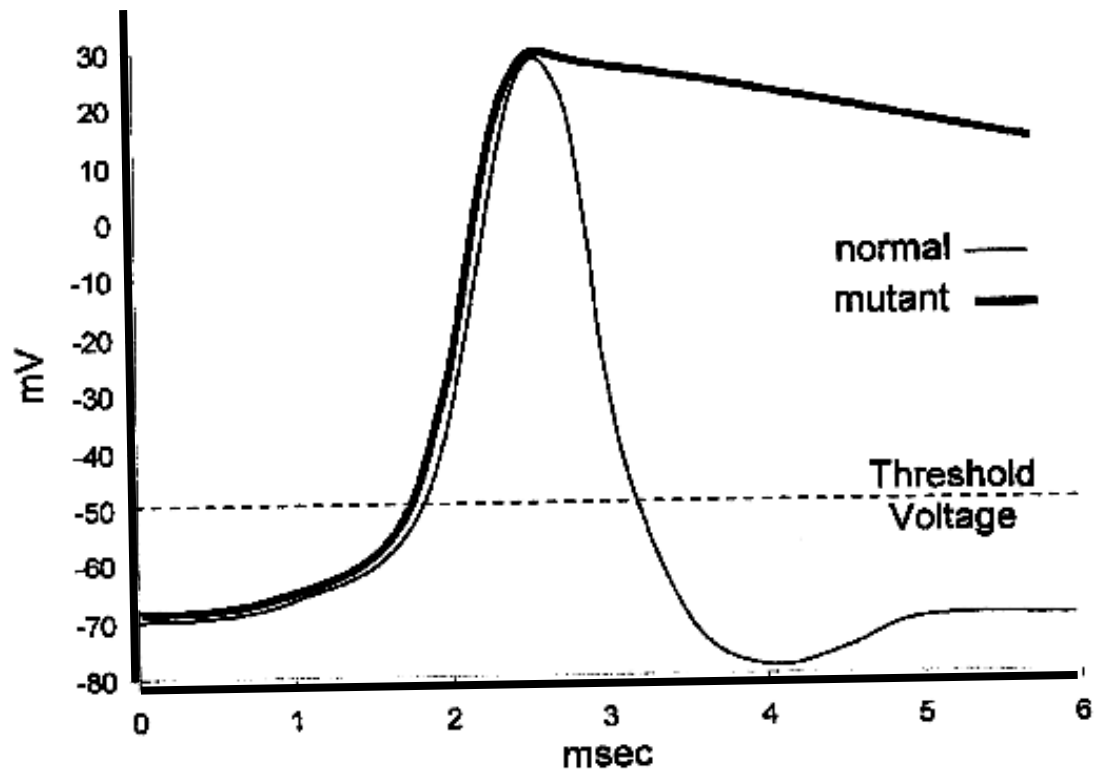
(a) Describe the events that take place after the arrival of the pre-synaptic action potential that results in the activation of dopamine receptors on the post-synaptic membrane. [3]

- **Depolarisation of pre-synaptic membrane causes voltage-gated calcium ion channels to open, resulting in Ca^{2+} influx down its electrochemical gradient**
- **Calcium ions cause the synaptic vesicles containing dopamine to move along cytoskeleton towards pre-synaptic membrane to fuse with membrane and to release dopamine via exocytosis**
- **Dopamine diffuses across synaptic cleft and binds to dopamine receptors on post-synaptic membrane → changes the conformation of receptors and activates the ligand gated Na^+ channels to open**

(b) State **two** reasons why nerve impulses can only travel in one direction across a synapse. [2]

- **Vesicles contain dopamine is only found in/dopamine is only synthesised at the pre-synaptic neurone**
- **Dopamine receptors are found only on the post-synaptic membrane**

Fig. 5.2 shows the action potentials of a normal and mutant neurone. In the mutant neurone, a mutation in the gene coding for a channel protein results in a change to the action potential.



(c) Describe how the resting potential of the normal neurone is maintained at -70mV. [2]

- $\text{Na}^+\text{-K}^+$ pump actively transports 3 Na^+ out and 2 K^+ in $\rightarrow$ Net loss of positive charge for every pumping action
- Facilitated diffusion via Na^+ and K^+ leakage (ungated) channels, which causes membrane to be 20x more permeable to K^+ / less permeable to Na^+ $\rightarrow$ More K^+ leaves cell than Na^+ enters $\rightarrow$ net loss of positive charge from the neurone due to electrochemical gradients of K^+ ;
- Impermeability of membrane to organic anions (too large), which has a higher concentration inside neurone

(e) Describe the trends observed in **Fig 6.4** for **E** and **E3**. [2]

- As stimulus intensity increases, the frequency of action potential increases.
- In E2, a weaker stimulus results in 7 action potentials being produced.
- Compared to E3, whereby a stronger stimulus results in 9 action potentials for the same time period,
- peak / amplitude of each action potential is the same, at about +15 mV

(f)(i) State what you understand by the term 'threshold potential'. [1]

refers to the **potential in which a membrane must be depolarised / reached for an action potential to be triggered**, has a **value of about –55 mV**

(ii) With reference to the results shown in **Fig 6.4** and **Table 6.1**, explain how stimulus intensity affects the extent of muscular contraction. [3]

- **In E1, a weak stimulus that produces a graded potential that is below threshold thus will not trigger an action potential.**
- **No neurotransmitters will be released into the synaptic cleft to cause muscle twitches.**
- **A strong stimulus causes the membrane potential to reach threshold, triggering a wave of action potentials which will result in the release of neurotransmitters to cause twitches.**
- **in E3, 11 contractions/APs are produced per second as compared to only 7 in E2 since a stronger stimulus is applied to the neurone in E3, resulting in a greater release of neurotransmitter → stronger and more sustained muscular contractions.**

Question 7

a) (i) State which genus does the orang-utan belong? [1]

- ***Pongo*;**

(ii) State which order does the chimpanzee belong? [1]

- **Primates / Primata;**

(b) Explain why this classification is both hierarchical and phylogenetic. [2]

- **Hierarchical because each level of category names is contained within a more inclusive category;**
- **Species within genus, genera within family, etc.;**
- **Phylogenetic because the hierarchy of names indicates the degree of relatedness among organisms;**
- **Ranking within more inclusive taxa may reflect recency of common ancestry;**

(c) Use the information in the diagram, state the present-day species of finch evolved first on the Galapagos islands? Explain your answer. [2]

Finch species:

- ***Certhidea olivacea*;;**

Explanation:

- **Lineage leading to *Certhidea olivacea* diverged first from the original species of finch from South America;**
- **Other species diverged later from other ancestries;**

(d) Explain why evaluating the DNA base sequences can work out the evolutionary relationship between the species of finches. [2]

- **All known life based on nucleic acids;**
- **Some genes / certain regions of genomes appear to evolve at constant rates;**
- **Extent of changes to DNA base sequences;**
- **Determine the extent of relatedness;**

(a) (i) Suggest how these different species of finch might have evolved from one species. [3]

- **Variation in beak shape present in original population;**
- **Selection of individuals with beaks suitable for eating small seeds on one island/selection of beak suitable for eating large seeds on another;**
- **Those with the most suited beaks (survived) to reproduce/pass on their alleles/genes for beak shape;**
- **Allele frequency changed in each gene pool on each island / populations became genetically different;**
- **No gene flow between islands/geographical isolation;**
- **So that individuals from different islands unable to reproduce to produce fertile offspring;**

(ii) Describe how biogeography may provide support for the evolution of the different species of finch. [2]

- **Biogeography is the geographic distribution of species;**
- **Species are found where they are because they evolved from ancestors that inhabited those regions;**
- **The Galapagos are colonized by finch species from the nearest mainland, South America;**
- **They adapt to their new environments and successful species best suited for the environment thrives and survives long enough to reproduce to pass on the gene to subsequent generations.**

Section B

Answer **only ONE** question.

Write your answer on the foolscap paper provided.

Question 8

(a)

E

Explain the role of the F plasmids in bacterial conjugation and suggest the advantages of conjugation over transduction. [8]

Explain the role of the F plasmids in bacterial conjugation: (max 4 marks)

- 1 Conjugation - transfer of genetic material (in one direction) through (direct) cell to cell contact between two bacterial cells;
- 2 Ref to one with the F plasmid (donor) and the other without (recipient);
- 3 F plasmid contains genes / codes for synthesis of sex pilus;
- 4 Used for attachment to recipient cell / formation of mating bridge/conjugation tube;
- 5 Ref. to F factor replicates by rolling-circle mechanism (*idea of replication is impt*);
- 6 (single-strand of) F plasmid is transmitted through mating bridge to recipient;
- 7 transferred ssDNA is replicated in the recipient cell, producing a ds F plasmid;
- 8 Conjugation tube breaks;
- 9 (if entire F plasmid was transferred) Recipient cell now has F plasmid/ is F⁺;
- 10 F plasmid can integrate into bacterial chromosome, forming Hfr cell;
- 11 Some genes from host chromosome are also transferred to recipient;
- 12 conjugation tube usually breaks before the entire F plasmid is transferred;

Suggest the advantages of conjugation as compared to transduction [max 4]

- 1 No need for 3rd party / virus / vice versa;
- 2 Host cell remains alive;
- 3 Does not rely on chance that wrong DNA fragments are packaged (in generalised transduction);
- 4 Does not rely on chance that viral DNA excises imprecisely (in specialised transduction);
- 5 Homologous recombination not needed unlike (general) transduction (though specialised transduction does not require it);
- 6 Higher chance to transfer genetic material (if cells contact each other) (as transduction relies heavily on chance);
- 7 In transduction, genetic material may or may not be degraded / vice versa;
- 8 Larger chunk of DNA/genetic material can be transferred;
- 9 Ref. defined genes (of F plasmid) transmitted during conjugation vs random DNA fragments (via general transduction) OR only genes/DNA fragments beside prophage (via specialised transduction) @ 1 mk

(b)

D

Differentiate between the viral genome and the eukaryotic genome. [7]

	Viral genome	Eukaryotic nuclear genome
Genetic material	1 Either DNA;	3 DNA;
	2 or RNA;	
	4 Either double-stranded;	6 Double-stranded;
	5 or single-stranded;	
	7 Either linear;	9 Linear;

	8 or circular ;	
	10 Either segmented; 11 or single;	12 segmented / many different chromosomes;
Presence of telomeres	13 May or may not be present;	14 Present
Presence of origin of replication	15 May or may not be present;	16 Yes / many present
DNA enclosed in ...	17 capsid;	18 nucleus;
DNA associated with histones	19 No;	20 Yes;
Amount of noncoding DNA	21 Little ;	22 Large amounts;
Amount of repetitive DNA / centromeres	23 Little / Absent	24 Large amounts/ Present
spacer DNA between genes	25 Little	26 Large amounts
Presence of introns in genes	27 No (in most cases);	28 Yes;
Size	29 Smaller;	30 Larger;

@ ½ mk each, award if attempt at valid comparison

(c) Describe the differences in the life cycle of T4 phage and HIV virus. [5]

Feature of comparison	T4 phage	HIV
Host cell	1 Infects bacterial cells	Infects human T-cells
Penetration / Entry	2 Phage does not undergo fusion with host cell's plasma membrane / contracts its tail sheath;	HIV envelope fuses with the host cell's plasma membrane;
	3 Phage injects only its ds DNA;	HIV releases its nucleocapsid (ssRNA and reverse transcriptase) into cytoplasm;
Uncoating	4 No uncoating required as capsid does not enter host cell;	Uncoating of nucleocapsid to release genome and enzymes;
Fate of viral nucleic acids	5 No reverse transcription / ds DNA is either immediately used as template for synthesis of viral proteins and nucleic acids (lytic cycle) or integrated into bacterial chromosome;	Its ssRNA is converted to dsDNA, using reverse transcriptase;
	6 Can enter lytic phase or lysogenic phase (prophage);	Viral genome incorporated into host cell chromosomes (provirus);
	7 Prophage is excised from host cell's chromosome upon spontaneous induction;	Provirus is not excised as viral mRNA is transcribed from viral DNA together with host cell's genes;

Integration of viral glycoproteins into cell membrane	8 No integration of viral glycoproteins into host's cell membrane;	Integration of viral glycoproteins (gp120 and gp41) into host's cell membrane;
Release / Exit	9 Host cell is lysed to release the new viruses;	The new viruses bud off from host cell's plasma membrane;

Question 9

(a) Describe the structural features of a named membranous organelle that is involved in ATP synthesis in photosynthesis. [6]

1. chloroplast

2. large / long / elongated / lens-shaped / 5 – 10 µm in length

3. enclosed by a double membrane / inner and outer membrane

4. inner membrane is selectively permeable containing protein channels

5. outer membrane is permeable to most ions especially H⁺ and molecules

6. contain gel-like aqueous matrix known as stroma

7. starch grains / ribosomes / circular DNA / enzymes of Calvin cycle present

8. flattened, disc-like sacs known as thylakoids suspended in stroma

9. arranged in stacks known as grana

10. which are joined by intergranal lamellae

11. thylakoid membrane 12. contains photosystems

12. photosynthetic pigments e.g. chlorophyll and carotenoids are present

13. electron carriers / electron transport chains are also embedded in the membrane

14. NADP⁺ reductase present

15. ATP synthase complex / stalked particles also present

16. fluid-filled interior space known as thylakoid lumen / space

- (b) Compare the two main pathways in which an excited electron can undergo during the light-dependent stage of photosynthesis. [6]

Differences:

Features	<i>non-cyclic photophosphorylation</i>	<i>cyclic photophosphorylation</i>
Photosystems involved	involves both PSII and PSI	involves only PSI
First electron donor (source of electrons)	from water , photolysis of water occurs	from P700 in PSI , no need water s.t. photolysis of water does not occur
Electron transport	involves plastoquinone, cytochrome complex, plastocyanin and ferredoxin . Also involves two electron transport chains .	involves ferredoxin, cytochrome complex and plastocyanin only & involves only one electron transport chain
Pathway of electron	non-cyclic , electron does not return to same molecule	cyclic , electron returns to same molecule
Final electron acceptor	NADP⁺	P700 in PSI
Generation of proton gradient	proton gradient generated by H ⁺ released during photolysis of water , and H ⁺ that are actively pumped from the stroma, across the thylakoid membrane, into the thylakoid lumen by energy released during electron flow .	proton gradient is generated by H ⁺ that are actively pumped from the stroma, across the thylakoid membrane, into the thylakoid lumen by energy released during electron flow
Products	ATP, NADPH & by-product O₂ produced	ATP only, no O₂ produced

Similarities:

- both pathways **occur at the thylakoid membrane**
- both pathways involve **PSI**
- both involve the **synthesis of ATP**
- both pathways uses **energy released during electron flow** to actively **pump H⁺** from the stroma, across the thylakoid membrane, **into the thylakoid lumen to produce a proton gradient** which drives ATP synthesis / involves **chemiosmosis**

- (c) With reference to the term 'limiting factor', discuss how two named factors have effects on photosynthesis. [6]

1. rate of photosynthesis is limited by the slowest reaction in the series
2. and when the rate of photosynthesis is limited by more than one factors
3. the rate is limited by the factor which is in the shortest supply
4. it determines the rate of reaction
5. light intensity
6. directly affects light-dependent stage of photosynthesis
7. at low light intensities, light intensity is the limiting factor
8. rate of photosynthesis increases proportionally with increasing light intensity
9. at high light intensities, rate of photosynthesis decreases
10. some other factor becomes the limiting factor
11. further increase in light intensity results in the light saturation point being reached
12. beyond which further increase in light intensity will have no effect on the rate of photosynthesis
13. light intensity is no longer a limiting factor
14. rate of photosynthesis is maximum at this point
15. carbon dioxide
16. a major limiting factor since atmospheric CO₂ concentration is low
17. CO₂ is needed in the light-dependent stage / Calvin cycle
18. it is reduced to carbohydrate / sugar
19. it is fixed by combining with ribulose-1,5-bisphosphate (RuBP)
20. increase in CO₂ concentration results in a proportional increase in the rate of photosynthesis
21. temperature
22. affects mainly light-independent stage and, to a certain extent, the light-dependent stage
23. are enzyme-controlled

- 24. increase in temperature, increases rate of photosynthesis
- 25. rate of photosynthesis doubles for each rise of 10°C until optimum temperature is reached
- 26. rate of photosynthesis decreases at higher temperature
- 27. as enzymes start to denature
- 28. oxygen
- 29. competes with CO₂
- 30. for the active site of the enzyme RuBP carboxylase oxygenase / Rubisco
- 31. during the light-independent stage / Calvin cycle
- 32. high concentration of oxygen decreases the rate of photosynthesis